

FlowGuard — MS1 Proposal

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1. Problem statement

FlowGuard predicts each station's water level 24 hours out, in centimeters, across 6 BAFU gauges in canton Uri. The cluster runs down the Reuss through Andermatt (2087), Erstfeld (2299) and Seedorf (2056), plus three gauges on the Schächen tributary near Bürglen: Galgenwäldli (2491), EW Altdorf (2492) and KW Schächen (2649). Each prediction draws on that station's own trailing 7 days of discharge (Q), level (W) and temperature (WT), plus the same window from its upstream neighbor where one exists. Nationwide extension (~310 stations) is a stretch goal; Uri is the committed MS1/MS2 scope.

Success: MAE ≤ 3 cm per station, beating a persistence baseline (tomorrow = today). Each prediction also maps to a risk band (low flow/drought, normal, flood risk) from that station's own 10th/90th percentile, which drives the dashboard. Flood/drought bands are rare versus normal, so I'll track precision/recall per band, not just overall accuracy.

2. Originality & motivation

Switzerland's 2026 drought brought record low river and lake levels alongside the flood risk that usually gets the attention; most forecasting projects, including past ones in this course, score only one direction. FlowGuard scores both from the same series, and Uri makes it personal: I know the Reuss and the Schächen, and can sanity check predictions against what I actually know of these rivers. What's distinct: the trend based features and chronological split below, plus treating the 6 stations as one connected system instead of 6 independent sensors.

3. Data source & features

Source: BAFU's Hydrological Data Service (data.bafu.admin.ch), a free GraphQL API, no auth. Verified live on 30.09.2026 for all 6 stations; readings land within 9 to 19 minutes, with hourly/daily/monthly aggregates per station going back years, growing with every new reading.

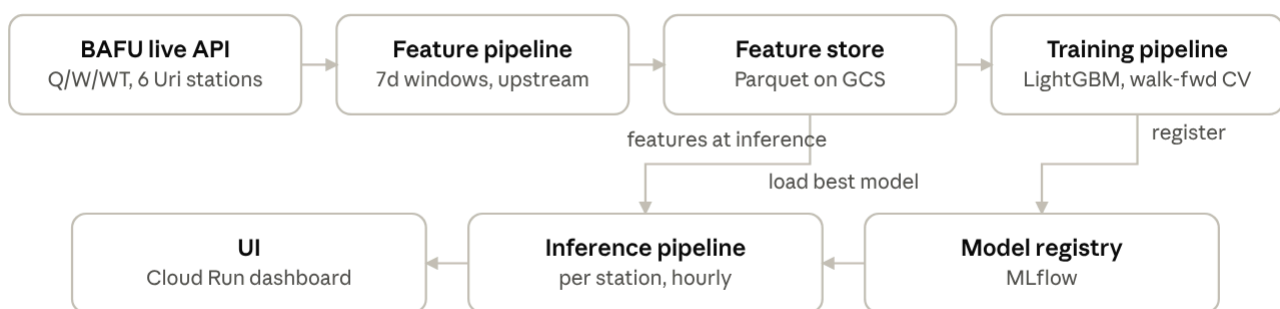
Features: each station's own water level 24h ahead is the label, a real future reading, not derivable from the inputs. Inputs use the trailing 7 days, not a single point, so there's a trend to learn from; a downstream station also gets the same features from its upstream neighbor:

- lags at $t-1h$, $-6h$, $-24h$, $-72h$, $-7d$
- rolling mean/std/min/max and rate of change, 24h and 72h windows
- the three above from the upstream neighbor too, where one exists
- station_id as a categorical feature, so one pooled model replaces 6 separate ones
- caveat: KW Schächen (2649) is a hydropower plant with managed flow

Split: strictly chronological across all 6 stations together, not per station, since that would leak information through the upstream features. Train on the oldest 70%, validate/test on the newest 30%, walk forward, never shuffled.

4. System design

Six Uri stations share one feature-to-inference pipeline



- **Feature pipeline** (GitHub Actions, hourly + backfill): pulls readings for all 6 stations, builds the features above, writes to the feature store.
- **Feature store:** versioned Parquet on GCS, partitioned by station and date, so nationwide scaling later is a config change, not a rewrite.
- **Training pipeline:** pooled LightGBM (station_id as a feature), tracked and registered with MLflow; preferred over an LSTM since the trend is already in the features.
- **Inference pipeline** (Cloud Run): loads the best model, reads each station's latest window, serves prediction and risk band via a dashboard.

Tech stack: GitHub, Docker, MLflow (tracking + registry), GitHub Actions (orchestration), GCS (feature store), Cloud Run (serving, \$50 GCP credit).

Stretch goals, not part of the core (core FTI ships first): nationwide scaling to all ~310 active BAFU stations, and MeteoSwiss precipitation as a second, causal source.